

Bilateral Automated Negotiation in Multi-Agent Systems

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Abstract

Automated negotiation is a central research area within multi-agent systems, that studies how autonomous agents reach agreements under conditions of incomplete information and conflicting preferences. This work provides a structured analysis of bilateral automated negotiation, with particular emphasis on time-dependent concession strategies, including Boulware, Conceder, and Linear.

Building upon this theoretical framework, we develop and implement a simulation environment to evaluate the performance of these strategies in a controlled and reproducible setting over randomly generated negotiation domains. The experimental analysis focuses on three key performance metrics: agreement rate, negotiation length, and social welfare.

The results reveal clear and systematic differences across strategies. In particular, mixed strategy interactions achieve higher agreement rates and social welfare, while homogeneous strategies such as Conceder vs Conceder lead to faster but lower-welfare outcomes.

These findings contribute to a better understanding of time-dependent negotiation dynamics and provide insights for the design of more effective autonomous negotiating agents. Finally, we discuss the limitations of the experimental framework and outline directions for future research.

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1 Introduction

Negotiation is a fundamental mechanism for conflict resolution in multi-agent systems, where autonomous agents with potentially conflicting preferences aim to reach mutually acceptable agreements. This paradigm has been extensively studied in artificial intelligence, particularly in domains such as electronic commerce, distributed resource allocation, and decision support systems [2, 3].

A central challenge in automated negotiation is the design of agent strategies that determine how preferences are revealed and how concessions are made over time. The concession policy adopted by an agent directly affects both the likelihood of reaching an agreement and the efficiency and quality of the negotiation outcome. Among the different approaches proposed in the literature, time-dependent strategies provide a widely used and analytically tractable framework for modeling concession behavior [5].

In time-dependent strategies, agents adjust their target utility over time, typically following predefined concession profiles such as Boulware, Conceder, or Linear. These strategies differ in the rate at which agents concede, ranging from highly conservative behaviors to more flexible and cooperative ones. Despite their simplicity, such strategies capture essential dynamics of negotiation processes and serve as a baseline for more sophisticated approaches [6].

In this work, we focus on the empirical evaluation of time-dependent strategies in bilateral negotiation settings. Specifically, we analyze how different concession profiles influence three key performance metrics: agreement rate, negotiation length, and social welfare. To this end, we implement a simulation framework in which agents employing different strategies engage in repeated negotiations over randomly generated domains.

The main contribution of this study is twofold. First, it provides a systematic experimental comparison of classical time-dependent strategies under controlled and reproducible conditions. Second, it provides quantitative insights into the trade-off between negotiation efficiency and outcome quality, and into how different concession profiles affect negotiation outcomes.

The remainder of this document is structured as follows. Section 2 introduces the basic concepts of automated negotiation. Section 3 presents a classification of automated negotiation settings. Sections 4 to 7 develop the main theoretical components of the bilateral negotiation framework, including protocols, strategies, utility models, and concession mechanisms. Section 8 describes the experimental setup, and Sections 8.2 and 8.3 present and discuss the experimental results.

2 Basic concepts of automated negotiation

We formally define automated negotiation as an interaction process among a set of autonomous agents that aim to reach a mutually acceptable agreement over a shared outcome space. Unlike centralized decision-making approaches, the outcome of the negotiation emerges from decentralized interactions among agents, each pursuing its own objectives [2, 3].

In this work, we focus on bilateral automated negotiation under incomplete information, where each agent has access only to its own preferences and must make decisions based on the observed behavior of its opponent.

2.1 Elements of a negotiation problem

We represent a negotiation problem as a tuple:

$$\mathcal{N} = \langle A, \Omega, U, r, P \rangle$$

where:

- $A = \{a_1, a_2\}$ is the set of negotiating agents,
- Ω is the outcome space (set of all possible agreements),
- $U = \{U_1, U_2\}$ is the set of utility functions, where $U_i : \Omega \rightarrow \mathbb{R}$ represents the preferences of agent a_i ,
- $r = \{r_1, r_2\}$ is the set of disagreement payoffs (also known as reservation values),
- P is the negotiation protocol governing the interaction.

In this work, however, we assume disagreement payoffs equal to zero. Therefore, reservation values do not affect the experimental decision process, although they remain relevant as a theoretical concept.

We typically define the outcome space Ω as the Cartesian product of issue domains:

$$\Omega = \prod_{i=1}^n D_i,$$

where each D_i represents the set of possible values for issue i . For example, in an e-commerce negotiation, the issues may correspond to price, delivery time, and warranty conditions.

2.2 Preference representation and utility

We commonly represent agent preferences using utility functions, which provide a numerical evaluation of each possible agreement:

$$U : \Omega \rightarrow \mathbb{R}.$$

Agents are assumed to be rational, meaning that they prefer outcomes with higher utility values. Utilities are often normalized to the interval $[0, 1]$ to facilitate comparison and simplify strategy design [1].

A key concept is the reservation value. Let r_i denote the utility that agent a_i receives if no agreement is reached. By definition, r_i is the disagreement payoff of the agent.

In theoretical models, a rational agent would never accept or propose an agreement o such that

$$U_i(o) < r_i.$$

Therefore, an agreement is individually rational for agent a_i only if

$$U_i(o) \geq r_i.$$

This condition defines the feasible set of agreements for each agent and plays a central role in theoretical models of negotiation.

2.3 Bilateral negotiation

In bilateral automated negotiation, two agents iteratively exchange proposals in order to reach an agreement.

The negotiation can be modeled as a discrete-time process:

$$t = 0, 1, 2, \dots, T,$$

where T denotes the negotiation deadline.

At each round:

- one agent proposes an offer $o_t \in \Omega$,
- the other agent decides whether to accept or reject it.

If the offer is accepted, the negotiation terminates with agreement. Otherwise, the process continues until either an agreement is reached or the deadline expires.

This interaction defines a sequential decision-making problem under incomplete information, where agents must balance the trade-off between achieving high utility and ensuring agreement [2].

2.4 Negotiation protocols

A negotiation protocol specifies the rules governing the interaction between agents. In this work, we adopt the alternating offers protocol, one of the most widely used mechanisms in bilateral negotiation [3].

Under this protocol, agents take turns proposing offers until one is accepted or the deadline is reached. In the absence of agreement, agents receive a disagreement payoff (also known as reservation value), which is the value received by the agent when no agreement is reached.

A more detailed description of negotiation protocols is provided in Section 4.

2.5 Negotiation strategies

A negotiation strategy defines how an agent selects its actions throughout the negotiation process. Formally, a strategy for agent a_i can be defined as a function:

$$\sigma_i : S \rightarrow \text{Act},$$

where S denotes the set of possible negotiation states and Act the set of available actions.

In practice, strategies determine:

- how offers are generated,
- when offers are accepted.

A common approach is to define strategies in terms of an aspiration function $\alpha(t)$, which specifies the target utility that the agent aims to achieve as a function of time.

The decision threshold, $\tau(t)$, is determined by the aspiration function $\alpha(t)$.

In theoretical negotiation models, this strategic threshold is subject to the additional constraint of individual rationality, which implies that an agent should not accept outcomes below its reservation value. In our experimental setting, we introduce a fixed minimum utility threshold, denoted as θ , which is independent from the reservation value. This threshold is used as a behavioral constraint for the agent's decisions. (see Section 8).

A detailed treatment of negotiation strategies is provided in Sections 5 and 7.

3 Types and classification of automated negotiation

Automated negotiation settings can be characterized along several orthogonal dimensions, including the number of agents, the structure of the outcome space, the temporal constraints, the information available to participants, and the nature of their preferences. These dimensions determine both the complexity of the negotiation process and the suitability of different strategy classes [4, 6].

In this section, we present a structured classification of automated negotiation scenarios, with particular emphasis on those most relevant to bilateral, multi-issue negotiation under incomplete information, which is the focus of this work.

3.1 Classification by number of agents

A primary distinction concerns the cardinality of the agent set A :

- **Bilateral negotiation:** $|A| = 2$. This is the simplest and most extensively studied setting, as it allows for precise analysis of strategic interaction and concession behavior. Many theoretical results and strategy designs are derived under this assumption.
- **Multilateral negotiation:** $|A| > 2$. In this case, the negotiation process becomes significantly more complex due to the presence of multiple potential agreement partners, coalition formation, and competing offers. The outcome space and the strategic space both expand considerably.

In this work, we restrict our analysis to bilateral negotiation, which provides a controlled environment for studying the impact of concession strategies.

3.2 Classification by number of issues

Another fundamental dimension is the structure of the outcome space Ω , determined by the number of issues:

- **Single-issue negotiation:** the outcome space is one-dimensional, i.e., $\Omega = D$, where D is a finite or continuous set. This setting is analytically tractable but limited in its ability to capture complex preference structures.
- **Multi-issue negotiation:** the outcome space is defined as a Cartesian product of multiple issue domains, where the number of issues satisfies $n \geq 2$. This structure enables agents to perform trade-offs across issues, increasing the likelihood of Pareto-efficient agreements [5].

Multi-issue negotiation is particularly relevant in practical applications and constitutes the setting adopted in this study.

3.3 Classification by temporal horizon

The role of time in negotiation is captured by the definition of a time horizon T :

- **Open-ended negotiation:** no fixed deadline is imposed ($T \rightarrow \infty$). Agents may continue negotiating indefinitely, which may lead to inefficient delays in reaching agreement.
- **Deadline-constrained negotiation:** the negotiation must terminate at or before a finite time T . If no agreement is reached within this horizon, the negotiation ends in disagreement.

The presence of a deadline introduces time pressure and is a key driver of concession behavior, particularly in time-dependent strategies [5].

3.4 Classification by available information

Negotiation settings can also be classified according to the information available to agents about their opponents:

- **Complete information:** each agent has full knowledge of the utility functions and strategies of other agents. This assumption is common in theoretical models but rarely holds in practice.
- **Incomplete information:** each agent has access only to its own utility function and must infer the preferences and behavior of its opponent from observed actions during the negotiation.

Most real-world negotiation scenarios, as well as the experimental setup considered in this work, fall under incomplete information, which introduces uncertainty and increases strategic complexity [6].

3.5 Classification by preference structure

Another relevant dimension concerns the relationship between agents' utility functions. Let U_1 and U_2 denote the utilities of two agents over Ω :

- **Zero-sum negotiation:** for all outcomes $o \in \Omega$,

$$U_1(o) + U_2(o) = c,$$

where c is a constant. In this case, one agent's gain is exactly the other's loss.

- **Non-zero-sum negotiation:** there exist outcomes $o, o' \in \Omega$ such that

$$U_1(o) + U_2(o) \neq U_1(o') + U_2(o'),$$

allowing for mutually beneficial agreements and Pareto improvements.

Most automated negotiation problems of practical interest are non-zero-sum, as they allow agents to exploit differences in preferences to achieve higher joint utility.

3.6 Distributive and integrative negotiation

Beyond the structural properties of the problem, negotiation can also be classified according to the strategic approach adopted by agents:

- **Distributive negotiation:** agents treat the negotiation as a process of dividing a fixed resource, focusing on maximizing their individual utility without considering potential gains from trade-offs.
- **Integrative negotiation:** agents actively search for agreements that improve the joint outcome by exploiting differences in preferences across issues.

It is important to note that this distinction refers to agent behavior rather than the intrinsic structure of the negotiation problem. In particular, even in non-zero-sum environments, agents may behave in a distributive manner if they fail to exploit mutually beneficial opportunities.

Multi-issue negotiation environments are especially conducive to integrative strategies, as they provide a richer structure for achieving efficient agreements through coordinated concessions and trade-offs.

4 Bilateral negotiation protocols

In automated negotiation, the protocol determines the interaction structure within which agents operate. While Section 2 introduced a formal definition of negotiation protocols, this section focuses on their practical instantiation in bilateral settings, emphasizing their role in shaping agent behavior and experimental design.

In bilateral negotiation, protocols define how agents exchange offers over time, how responses are handled, and under which conditions the interaction terminates. These elements directly constrain the space of admissible strategies and strongly influence negotiation outcomes.

4.1 Protocol components

We can characterize a bilateral negotiation protocol by a set of operational components that determine the dynamics of interaction:

- **Turn-taking mechanism:** specifies which agent acts at each round. In most bilateral settings, agents alternate deterministically.
- **Communication primitives:** defines the set of admissible actions, typically including proposal, acceptance, and implicit rejection through counteroffers.
- **Offer representation:** agreements are represented as vectors in the outcome space, where each component corresponds to a value of a negotiated issue.
- **Admissibility constraints:** restrict the set of valid offers according to the domain definition (e.g., issue domains and consistency conditions).
- **Termination conditions:** specify when the negotiation ends, either due to agreement or the expiration of a predefined time horizon.

These components determine the structure of the negotiation process and define the environment in which agent strategies are executed.

4.2 Alternating offers protocol

The alternating offers protocol is the standard interaction mechanism in bilateral automated negotiation [3]. Under this protocol, agents take turns proposing offers until one is accepted or a termination condition is reached.

At each round, the current proposer selects an offer from the outcome space, and the responding agent evaluates it according to its decision rule. If the offer is accepted, the negotiation terminates with agreement; otherwise, the responding agent becomes the proposer in the next round.

This protocol is particularly suitable for negotiation under incomplete information, as it allows agents to infer partial information about their opponent's preferences from observed proposals and responses.

Figure 1 illustrates the sequential exchange of offers under this protocol.

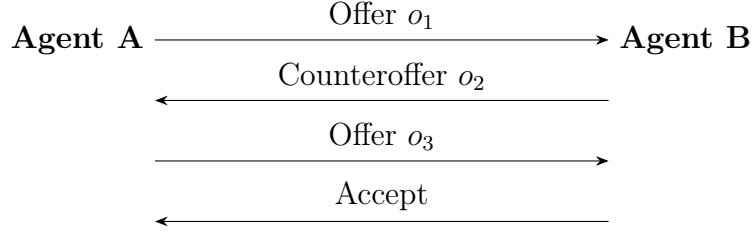


Figure 1: Sequential interaction in the alternating offers protocol. Agents alternate roles as proposer and responder until agreement or termination.

4.3 Communication acts

In practice, bilateral negotiation protocols rely on a minimal set of communicative actions:

- **Propose:** the agent submits a concrete offer $o \in \Omega$.
- **Accept:** the agent accepts the last received offer, terminating the negotiation.
- **Counteroffer:** a new proposal implicitly rejects the previous one and continues the negotiation.

These primitives are sufficient to model most bilateral negotiation processes. Additional messages, such as session initialization or information exchange, are typically handled at the system level and do not directly influence strategic decision-making.

4.4 Termination and time constraints

In many negotiation scenarios, including the one considered in this work, the protocol includes a finite time horizon or deadline. If no agreement is reached before this deadline, the negotiation terminates without agreement, and agents obtain their disagreement outcome.

The presence of a deadline has a direct impact on agent behavior. In particular, it introduces a trade-off between maximizing utility and ensuring agreement, as agents must decide how long to maintain high aspiration levels before conceding.

This time pressure is a central factor underlying time-dependent strategies, which adjust concession behavior as the negotiation approaches its deadline.

5 Agent-based negotiation strategies

Given a negotiation protocol, the behavior of agents is fully determined by the strategies they employ. Building upon the formal definition introduced in Section 2, this section focuses on the main qualitative properties and design dimensions of negotiation strategies in automated settings [3, 6].

In automated negotiation, strategies determine how agents generate offers and evaluate incoming proposals over time. Their design has a direct impact on both the probability of reaching agreement and the efficiency and quality of the negotiated outcome.

The concepts presented in this section are general and independent of the specific implementation. The concrete strategies used in the experimental evaluation are described in Sections 7 and 8.

5.1 Objectives of negotiation strategies

The primary objective of a negotiation strategy is to maximize the expected utility of the agent. However, in deadline-constrained negotiations, this objective is subject to additional constraints imposed by time and uncertainty.

In particular, agents must balance two competing goals:

- **Outcome optimality:** achieving agreements with high utility,
- **Agreement probability:** ensuring that negotiation concludes successfully before the deadline.

This trade-off is a central element in the design of negotiation strategies and is widely discussed in the literature [2, 3].

5.2 Deterministic and stochastic strategies

We can classify strategies according to the role of randomness in their decision-making process:

- **Deterministic strategies:** the agent's actions are fully determined by the current negotiation state.
- **Stochastic strategies:** decisions incorporate probabilistic elements, allowing the agent to randomize among possible actions.

Stochastic strategies are often used to increase robustness and reduce predictability, particularly in environments where opponents may attempt to exploit deterministic behavior [6].

In this work, we focus on deterministic strategies, which provide a controlled and reproducible setting for experimental analysis.

5.3 Reactive and proactive behavior

Another relevant distinction concerns the extent to which agents actively influence the negotiation process:

- **Reactive behavior:** the agent focuses on evaluating and responding to incoming offers based on predefined criteria.
- **Proactive behavior:** the agent actively generates offers to guide the negotiation toward favorable regions of the outcome space.

In most practical implementations, negotiation strategies combine both behaviors, alternating between proposing offers and responding to the opponent's actions [3].

5.4 Time dependence in strategies

Time plays a central role in many negotiation strategies, particularly in settings with a finite deadline. As the negotiation progresses, agents typically adjust their behavior in order to avoid reaching the deadline without agreement.

This temporal adaptation is commonly implemented through time-dependent decision rules, in which agents progressively relax their acceptance thresholds and modify their offer generation criteria [5].

Such mechanisms provide a principled way to manage the trade-off between maintaining high utility and ensuring agreement.

5.5 Opponent modeling and adaptation

Strategies can also be classified according to whether they incorporate information about the opponent:

- **Opponent-independent strategies:** the agent makes decisions based solely on its own preferences and the current negotiation state.

- **Adaptive strategies:** the agent updates its behavior based on observed offers, attempting to infer the opponent’s preferences or strategy.

Adaptive strategies can improve performance by exploiting opponent-specific information, but they introduce additional complexity and may be sensitive to estimation errors. In contrast, opponent-independent strategies are typically simpler and more robust, making them suitable as baseline approaches [6].

In the experimental framework considered in this work, agents do not perform opponent modeling and follow opponent-independent strategies, which allows us to isolate the effect of time-dependent concession behavior.

5.6 Design trade-offs

The design of negotiation strategies involves several trade-offs between simplicity, robustness, and performance. Simple strategies are easier to implement and analyze but may fail to exploit opportunities for improved outcomes. More sophisticated strategies can achieve better performance in specific environments but often require additional information and may be less stable under uncertainty.

Consequently, the effectiveness of a strategy depends strongly on the negotiation context, including the protocol, the time horizon, and the information available to the agents [2, 3].

6 Preference representation and utility functions

The ability to evaluate and compare possible agreements is a fundamental requirement for autonomous negotiation. While the basic definition of utility functions has been introduced previously, this section focuses on multi-attribute utility models, normalization, Pareto efficiency, and their implications for negotiation outcomes [1, 6].

6.1 Preferences in automated negotiation

An agent’s preferences define how desirable each possible outcome is. In most automated negotiation scenarios, preferences are private information, meaning that each agent has access only to its own utility function. This introduces uncertainty into the negotiation process and requires agents to make decisions based on incomplete information.

In multi-issue negotiation settings, preferences must capture the agent’s valuation over combinations of issue values. This increases the complexity of decision-making but also creates opportunities for mutually beneficial agreements through trade-offs across issues [5].

6.2 Utility functions

Utility functions provide a numerical representation of preferences, allowing agents to rank and compare different outcomes. Formally, a utility function assigns a real value to each possible agreement:

$$U : \Omega \rightarrow \mathbb{R}.$$

Higher utility values correspond to more preferred outcomes. This representation enables negotiation to be treated as an optimization problem, where agents aim to select actions that maximize their expected utility [2].

6.3 Utility normalization

Many negotiation frameworks normalize utility values to the interval $[0, 1]$. Normalization preserves the ordinal structure of preferences while facilitating comparison across agents and simplifying strategy design.

This assumption is particularly useful in computational settings, where strategies often rely on thresholds or aspiration levels defined over a fixed range [1].

6.4 Utility functions in multi-attribute negotiation

In multi-issue negotiation, a common modeling assumption is that total utility decomposes into a weighted sum of individual issue utilities. Under this additive model, the utility of an outcome is expressed as:

$$U(o) = \sum_{i=1}^n w_i v_i(o_i),$$

where w_i represents the importance of issue i , and $v_i(o_i)$ denotes the utility of the selected value for that issue.

This representation enables agents to perform trade-offs across issues and is widely used due to its simplicity and interpretability. However, it assumes independence between issues, which may not hold in more complex or realistic scenarios [5, 6].

6.5 Pareto efficiency in negotiation outcomes

When analyzing negotiation outcomes involving two agents, it is useful to consider the joint utility space. Each outcome can be represented as a point $(U_1(o), U_2(o))$.

An outcome o is said to be *Pareto efficient* if there is no other outcome o' such that:

$$U_1(o') \geq U_1(o), \quad U_2(o') \geq U_2(o),$$

with at least one strict inequality.

The set of all Pareto-efficient outcomes forms the Pareto frontier, which represents the boundary of feasible optimal agreements. Negotiation strategies often aim to reach outcomes close to this frontier, as they correspond to efficient allocations of utility [2].

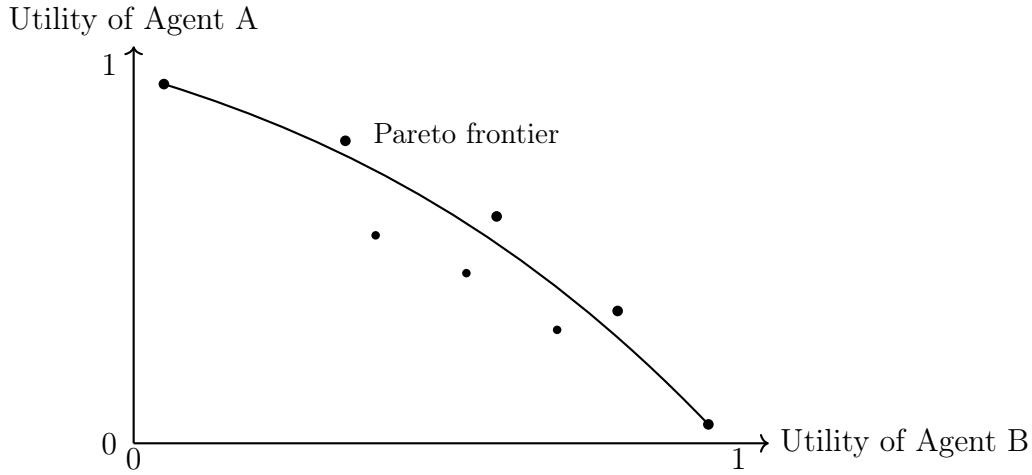


Figure 2: Example of a negotiation outcome space for two agents. Points on the Pareto frontier correspond to efficient agreements.

6.6 Individual rationality

A fundamental constraint in automated negotiation is individual rationality. Let r_i denote the reservation value of agent i . An agent will accept an outcome o only if:

$$U_i(o) \geq r_i.$$

This condition defines the set of acceptable agreements for each agent and plays a central role in both offer generation and acceptance decisions [1].

6.7 Limitations of utility-based models

Although utility functions provide a powerful and tractable representation of preferences, they also have limitations. In practice, specifying accurate utility functions can be difficult, especially in environments where preferences are incomplete, uncertain, or dynamic.

Furthermore, numerical representations may fail to capture qualitative or context-dependent aspects of decision-making. These limitations motivate the development of more advanced models incorporating learning, uncertainty, or richer preference structures [6].

7 Concession, aspiration, and acceptance criteria

In bilateral automated negotiation, agent behavior is governed by mechanisms that regulate how preferences evolve over time and how decisions are made in response to incoming offers. In particular, concession policies, aspiration levels, and acceptance criteria determine how agents balance utility maximization against the risk of negotiation failure.

This section provides a detailed treatment of the strategy components introduced in Section 2.

7.1 Concession mechanisms

Concession refers to the process by which an agent progressively relaxes its requirements during negotiation. Let $t \in [0, 1]$ denote normalized negotiation time. A concession policy can be understood as a function that determines how the agent's acceptance threshold decreases over time.

In early stages of negotiation, agents typically maintain high thresholds, proposing offers close to their optimal outcomes. As time progresses, these thresholds are reduced to increase the likelihood of reaching agreement before the deadline.

Concession behavior is therefore a key factor in ensuring the viability of the negotiation process and has been extensively studied in the literature [5].

7.2 Aspiration levels

The aspiration level determines the agent's strategic threshold over time. Let $\alpha(t)$ denote the aspiration function of an agent, where $t \in [0, 1]$ is normalized negotiation time.

At early stages, $\alpha(t)$ is typically close to 1, reflecting highly demanding behavior. As t approaches 1, $\alpha(t)$ decreases, making the agent more flexible and increasing the probability of agreement.

In theoretical negotiation models, this strategic threshold is subject to the additional constraint of individual rationality, which implies that an agent should not accept outcomes below its reservation value. In our experimental setting, we introduce a fixed minimum utility threshold, denoted as θ , which is independent of the reservation value. This threshold acts as a behavioral constraint for the agent's decisions.

7.3 Time-dependent concession functions

Time-dependent strategies define the aspiration function explicitly as a function of time. In this work, we consider three canonical forms, consistent with the literature [5]:

- **Linear strategy:**

$$\alpha(t) = 1 - kt$$

- **Conceder strategy:**

$$\alpha(t) = 1 - kt^\beta, \quad 0 < \beta < 1$$

- **Boulware strategy:**

$$\alpha(t) = 1 - kt^\beta, \quad \beta > 1$$

These formulations are simplified canonical forms commonly used to represent different concession profiles over time.

Here, $k \in (0, 1]$ determines the concession range. These functions generate different concession profiles: Conceder strategies concede rapidly at early stages, Boulware strategies delay concessions until late in the negotiation, and Linear strategies exhibit constant concession speed.

Figure 3 illustrates representative concession patterns.

7.4 Acceptance criterion

Given an incoming offer $o \in \Omega$, the agent determines the acceptance decision by comparing its utility to the current threshold:

$$U_i(o) \geq \tau(t).$$

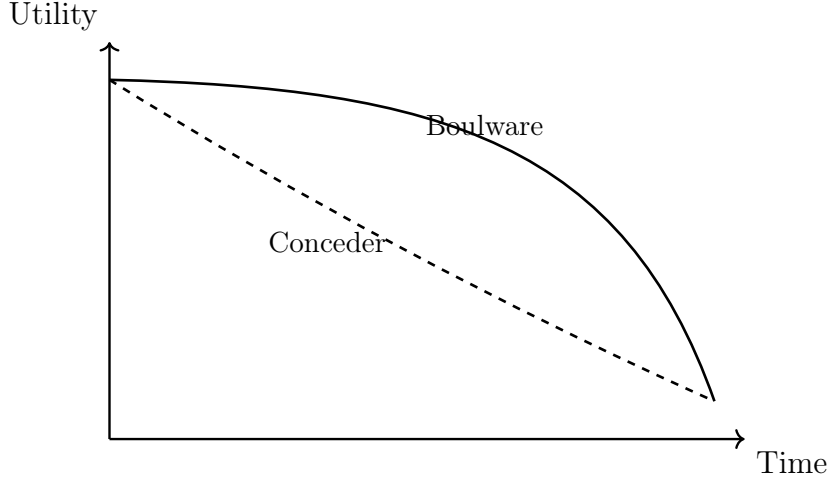


Figure 3: Representative time-dependent concession profiles.

If this condition is satisfied, the agent accepts the offer and the negotiation terminates. Otherwise, the agent continues the negotiation process.

This rule is consistent with the principle of individual rationality, as it enforces a minimum acceptable utility level through the decision threshold.

7.5 Offer generation under aspiration constraints

When acting as proposer, the agent selects an offer whose utility satisfies its current decision threshold.

Among all feasible offers, the agent selects the one with the lowest utility for itself. This choice ensures that concession behavior is reflected in the sequence of proposals, as the agent gradually proposes less favorable offers over time.

Thus, the decision threshold determines the set of admissible offers, while the selection rule determines which specific offer is proposed.

7.6 Trade-off between utility and agreement

The design of concession and acceptance mechanisms introduces a fundamental trade-off between maximizing utility and ensuring agreement.

Strategies that maintain high aspiration levels for too long may fail to reach agreement within the deadline. Conversely, strategies that concede too quickly may lead to suboptimal outcomes.

Time-dependent strategies provide a systematic way to navigate this trade-off by adjusting aspiration levels as a function of time, balancing efficiency and outcome quality [5].

8 Experimental evaluation

8.1 Experimental setup

To complement the theoretical analysis, we develop a simulation-based experimental framework to study the behavior of different negotiation strategies in a controlled and reproducible environment.

Negotiation domains. We automatically generate a total of 10,000 independent negotiation domains. Each domain is defined as a discrete multi-issue negotiation space with exactly three issues. For each issue, the number of available options is sampled uniformly at random between 3 and 5.

Formally, let m_i denote the number of values of issue i . The outcome space of each domain is defined as:

$$\Omega = \prod_{i=1}^3 \{0, 1, \dots, m_i - 1\}.$$

We explicitly enumerate all possible offers in each domain prior to the start of the negotiation.

Utility functions. For each domain, we assign each agent an independently generated additive utility function. Let $o = (o_1, o_2, o_3) \in \Omega$ denote an offer. The utility of o for an agent is defined as:

$$U(o) = \sum_{i=1}^3 w_i v_i(o_i),$$

where w_i is the weight associated with issue i , and $v_i(o_i)$ is the utility assigned to the selected value.

Weights and value utilities are randomly generated and normalized such that $\sum w_i = 1$ and each issue value lies in $[0, 1]$.

Negotiation protocol. All negotiations follow a bilateral alternating-offers protocol. Each negotiation involves two agents and lasts for at most $R = 12$ rounds.

At round $r \in \{0, 1, \dots, R - 1\}$, normalized time is defined as:

$$t = \frac{r}{R - 1}.$$

If an offer is accepted, the negotiation terminates with agreement. Otherwise,

agents alternate roles. If no agreement is reached within the deadline, the negotiation ends without agreement and both agents receive zero utility.

Offer generation and acceptance rule. Before negotiation starts, each agent enumerates all possible offers and sorts them in decreasing order of its own utility.

The decision threshold is defined as:

$$\tau(t) = \max\{\theta, \alpha(t)\},$$

where θ is a fixed minimum utility threshold and $\alpha(t)$ is the aspiration function.

When acting as proposer, the agent selects the offer with the **lowest utility for itself** among those satisfying:

$$U(o) \geq \tau(t).$$

This rule ensures that concession behavior is explicitly reflected in the sequence of proposals over time: as $\tau(t)$ decreases, the agent progressively proposes less favorable offers.

When acting as responder, the agent accepts an offer o if:

$$U(o) \geq \tau(t).$$

Strategies. We implement three time-dependent strategies:

- **Linear:**

$$\alpha_{\text{Linear}}(t) = 1 - 0.9t$$

- **Conceder:**

$$\alpha_{\text{Conceder}}(t) = 1 - 0.9t^{0.5}$$

- **Boulware:**

$$\alpha_{\text{Boulware}}(t) = 1 - 0.9t^3$$

Minimum utility threshold. We fix $\theta = 0.7$, which defines the minimum acceptable utility level for both proposal and acceptance decisions and is treated as a design parameter of the agents.

Experimental design. We evaluate all unordered pairs of strategies:

$$\{\text{Boulware}, \text{Conceder}, \text{Linear}\},$$

resulting in 6 combinations and 60,000 negotiation sessions.

Computational details. We executed the experiments on a personal computer running Windows 11 Pro, equipped with an 11th Gen Intel(R) Core(TM) i5-1155G7 processor at 2.50 GHz and 16 GB of RAM. Due to the simplicity of the negotiation model, the explicit enumeration of the relatively small outcome spaces, and the short negotiation horizon of 12 rounds, the full experiment required only about 20 seconds of execution time.

Evaluation metrics. We evaluate each strategy pair using three metrics:

- **Agreement rate:** the proportion of negotiation sessions that end in agreement.
- **Negotiation length:** the average number of rounds required until termination, either by agreement or by deadline expiration.
- **Social welfare:** the sum of the utilities obtained by both agents in the final agreement; if no agreement is reached, social welfare is zero.

8.2 Results

Table 1 reports the results.

Table 1: Experimental results. Agreement rate and social welfare values are reported in percentage form (multiplied by 100) for improved readability.

Strategy pair	Agreement rate (%)	Avg. length	Avg. welfare (%)
Boulware vs Boulware	77.74 ± 0.82	8.59 ± 0.05	132.46 ± 1.42
Boulware vs Conceder	82.57 ± 0.74	6.12 ± 0.07	144.47 ± 1.33
Boulware vs Linear	81.68 ± 0.76	6.79 ± 0.06	142.54 ± 1.35
Conceder vs Conceder	72.48 ± 0.88	5.20 ± 0.08	115.41 ± 1.41
Conceder vs Linear	77.85 ± 0.81	5.24 ± 0.07	130.05 ± 1.38
Linear vs Linear	77.96 ± 0.81	5.75 ± 0.07	130.55 ± 1.39

The results are illustrated in Figures 4, 5, and 6.

Figure 4 shows noticeable differences in agreement rates across strategy pairs. Combinations involving Boulware and Conceder achieve the highest agreement rates, while Conceder vs Conceder produces the lowest.

Figure 5 highlights clear differences in negotiation duration. Boulware vs Boulware results in the longest negotiations, whereas Conceder-based interactions converge significantly faster.

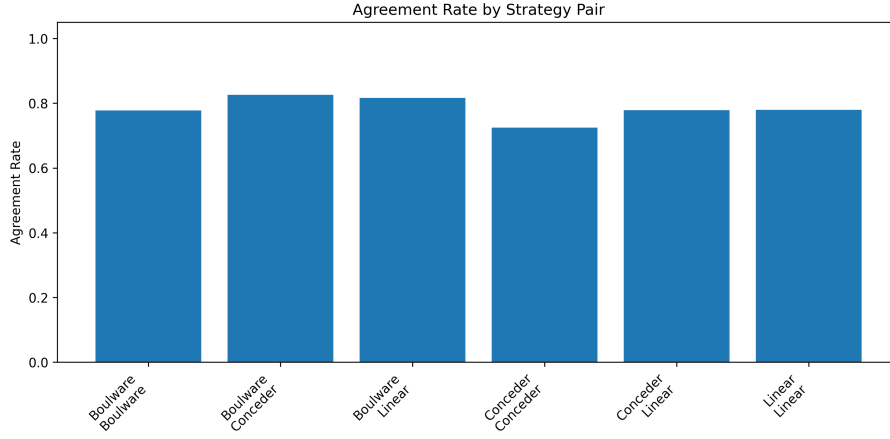


Figure 4: Agreement rate by strategy pair.

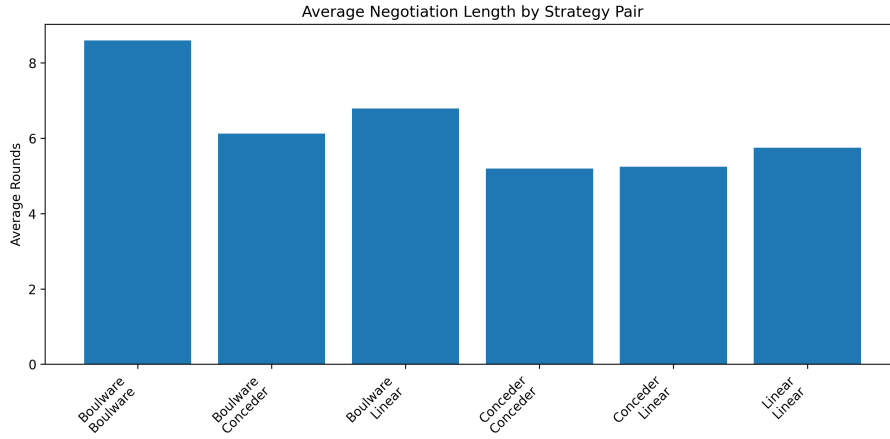


Figure 5: Average negotiation length by strategy pair.

Figure 6 reveals substantial differences in social welfare. The highest welfare is achieved in mixed strategy settings such as Boulware vs Conceder, while Conceder vs Conceder yields the lowest outcomes.

Statistical analysis. Each value corresponds to the mean over 10,000 runs. Confidence intervals are computed as:

$$CI_{95\%} = 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

The intervals are narrow, indicating stable estimates. Differences between strategy pairs are larger than the confidence intervals, suggesting statistically meaningful effects.

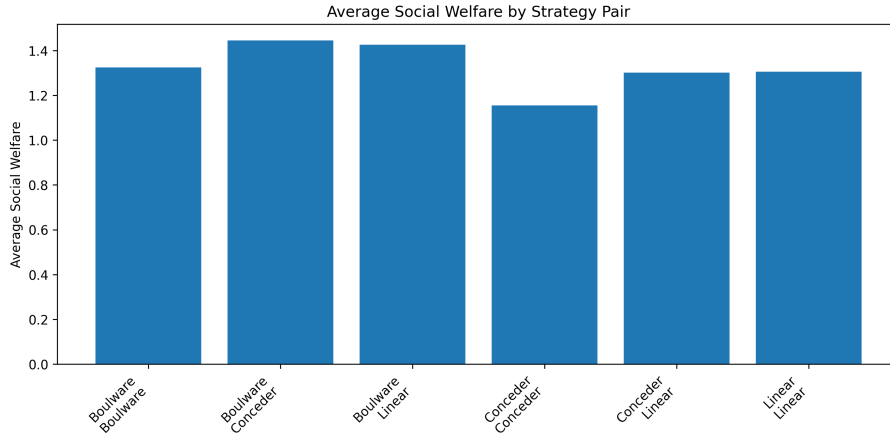


Figure 6: Average social welfare by strategy pair.

8.3 Discussion

The results reveal clear and systematic differences between negotiation strategies.

First, agreement rates vary across strategy combinations. The highest agreement rates are observed in asymmetric settings such as Boulware vs Conceder, suggesting that complementary concession behaviors facilitate agreement. In contrast, Conceder vs Conceder yields lower agreement rates, indicating that excessive early concessions do not necessarily improve success probability.

Second, negotiation length strongly depends on concession profiles. Boulware strategies delay concessions and therefore prolong negotiations, while Conceder strategies reach agreements significantly faster. Linear strategies exhibit intermediate behavior.

Third, social welfare differs substantially across strategies. The highest welfare is achieved in mixed settings such as Boulware vs Conceder, while Conceder vs Conceder produces the lowest welfare. This indicates that rapid concessions tend to reduce agreement quality, whereas more conservative strategies allow agents to reach more efficient outcomes.

Overall, the results highlight a clear trade-off between negotiation efficiency and outcome quality. Faster strategies achieve agreements more quickly but at lower utility levels, while slower strategies improve welfare at the cost of longer negotiations.

9 Negotiation in electronic commerce

Agent-based automated negotiation has found one of its main application domains in electronic commerce, where software agents act on behalf of individuals or orga-

nizations to autonomously negotiate the terms of transactions [3, 4]. These terms may include price, quantity, delivery time, or contractual conditions.

The use of negotiating agents enables the automation of decision-making processes in dynamic environments, reducing operational costs and increasing efficiency in digital markets.

9.1 Electronic markets and negotiating agents

Electronic markets are digital platforms where multiple buyers and sellers interact simultaneously. In these environments, agents may participate in auctions, bilateral negotiations, or hybrid mechanisms combining both approaches [4].

Each agent represents the preferences of its owner and applies negotiation strategies to reach utility-maximizing agreements. The presence of multiple self-interested agents with potentially conflicting objectives makes automated negotiation a central mechanism in these systems.

9.2 Bilateral vs. multilateral negotiation

While bilateral negotiation is common in direct buyer–seller interactions, many real-world electronic markets are inherently multilateral. In such settings, agents negotiate with multiple potential counterparts, often in parallel.

This introduces additional complexity, as agents must consider not only the negotiation process itself but also the availability of alternative agreements. As a result, competition among potential partners can significantly influence concession behavior and negotiation outcomes [2].

9.3 Multi-attribute negotiation

Electronic commerce transactions typically involve multiple attributes beyond price, such as quality, delivery conditions, or service guarantees. This leads to multi-attribute negotiation scenarios, where agents must evaluate complex offers and perform trade-offs across issues.

Utility-based models are commonly used in this context to aggregate preferences over multiple attributes, enabling agents to identify efficient agreements [5].

9.4 Applications and limitations

Automated negotiation has been applied in domains such as supply chain management, service-level agreement negotiation, and resource allocation in digital markets.

However, its adoption in real-world systems remains limited by several challenges.

These include difficulties in accurately modeling user preferences, handling incomplete information, and ensuring interoperability among heterogeneous agents. Additionally, issues such as trust, reputation, and strategic manipulation remain critical for practical deployment [6].

10 Current challenges and limitations of automated negotiation

Despite significant advances, the practical deployment of automated negotiation systems still faces important challenges. These arise from both computational limitations and the difficulty of modeling realistic agent behavior.

10.1 Realistic preference modeling

Many negotiation models assume that agent preferences are known, stable, and easily represented through utility functions. In practice, however, preferences may be incomplete, dynamic, or difficult to quantify.

This mismatch between theoretical assumptions and real-world behavior remains one of the main obstacles to practical adoption [6].

10.2 Incomplete and asymmetric information

Agents typically operate under incomplete information, lacking knowledge about the opponent's preferences or strategy. This informational asymmetry complicates decision-making and may lead to inefficient or unstable outcomes [2].

10.3 Computational complexity

In multi-issue or multilateral negotiations, the size of the outcome space can grow exponentially. This makes exhaustive search infeasible and requires the use of heuristics or approximation methods.

Designing efficient algorithms that operate under time constraints while maintaining good performance remains a challenging problem.

10.4 Robust strategy design

Negotiation agents must operate effectively against a wide range of opponents, from cooperative to highly competitive ones. Developing strategies that are robust across different environments without relying on strong assumptions is still an open research problem [3].

10.5 Trust and strategic behavior

In open systems, agents may behave strategically or opportunistically. Mechanisms for trust and reputation are therefore essential to ensure reliable interactions, but integrating these mechanisms into negotiation frameworks introduces additional complexity.

10.6 Bridging theory and practice

A major challenge is transferring theoretical models to real-world applications. While many approaches perform well in controlled environments, their effectiveness may degrade under real-world conditions, where factors such as uncertainty, regulation, and human interaction play a significant role.

11 Research directions and future trends

Agent-based automated negotiation remains an active research area, driven by both theoretical developments and the increasing complexity of real-world applications. Recent research efforts focus on extending classical models to operate effectively in dynamic, uncertain, and large-scale environments [6, 3].

One of the most prominent trends is the integration of machine learning techniques into negotiation processes. Instead of relying on predefined strategies, agents are increasingly designed to learn from interaction data, adapting their behavior over time. In particular, reinforcement learning and Bayesian approaches have been explored to enable opponent modeling and dynamic adjustment of concession policies.

Another important research direction concerns negotiation in open environments, where agents may enter or leave the system and where information about participants is incomplete or unreliable. These scenarios require robust mechanisms capable of operating under uncertainty, without strong assumptions about rationality or cooperation.

Multilateral and multipartite negotiation also constitute an important area of study. When more than two agents are involved, the negotiation space grows significantly and new challenges arise, such as coalition formation and coordination among multiple participants [2].

In addition, there is increasing interest in incorporating human and social factors into negotiation models. Concepts such as fairness, trust, and reputation are being integrated into agent design to improve realism and facilitate interaction with human users.

Finally, emerging technological developments, including large-scale digital markets and agents enhanced with advanced language models, open new possibilities for richer negotiation processes. These developments enable more expressive communication and more flexible interaction, but also introduce new challenges related to control, transparency, and alignment.

Overall, current research directions aim to develop negotiation systems that are more adaptive, robust, and aligned with real-world requirements.

12 Conclusions

This work presents a comprehensive analysis of agent-based automated negotiation, combining a formal theoretical framework with an extensive experimental evaluation of time-dependent negotiation strategies.

From a theoretical perspective, the study introduces the fundamental components of automated negotiation, including preference representation through utility functions, interaction protocols, and strategy design based on concession, aspiration, and acceptance mechanisms. These elements provide a structured foundation for understanding how autonomous agents operate in negotiation environments under incomplete information.

From an experimental perspective, we developed a large-scale simulation framework to evaluate the behavior of Linear, Conceder, and Boulware strategies across 60,000 independent negotiation sessions over randomly generated domains. The results reveal clear and systematic differences between strategies.

In particular, asymmetric combinations such as Boulware vs Conceder achieve the highest agreement rates and social welfare, indicating that complementary concession behaviors facilitate both successful and efficient outcomes. In contrast, homogeneous interactions such as Conceder vs Conceder lead to faster agreements but significantly lower social welfare, highlighting the cost of excessive early concessions.

These findings reveal a fundamental trade-off between negotiation efficiency and

outcome quality. Strategies that concede rapidly tend to reduce negotiation length but at the expense of outcome efficiency, while more conservative strategies improve welfare but prolong the negotiation process.

Overall, the results demonstrate that, under the considered setting, the choice of strategy plays a significant role in shaping negotiation outcomes.

The study further emphasizes that classical time-dependent strategies, while simple and analytically tractable, capture essential dynamics of negotiation processes. However, their limitations remain evident, particularly in the absence of opponent modeling and adaptive behavior, which restricts the strategic richness of interactions.

Finally, the large number of simulated negotiations ensures statistically reliable results, with narrow confidence intervals supporting the robustness of the observed patterns.

In conclusion, agent-based automated negotiation provides a powerful framework for distributed decision-making. The results of this study highlight the importance of concession dynamics and strategic interaction in determining both the efficiency and quality of negotiation outcomes. Future research should focus on incorporating adaptive mechanisms, opponent modeling, and more realistic preference structures to better capture the complexity of real-world negotiation environments.

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